

Subject-Disjoint Benchmarking of Nano Detectors for Visible Driver-Cue Localization

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Abstract—Object detection can localize visible evidence inside the cabin, but detection average precision does not establish a temporal driver state. We therefore benchmark single-frame localization of four discrete, momentary driver cues under a fixed subject-disjoint protocol. The primary RGB experiment retains all 15,723 sampled DMD frames from 14 drivers, including 12,722 negatives, and evaluates released YOLO11n, YOLO26n, and D-FINE-N systems across three predeclared seeds. Confidence selection uses validation only; protected testing reports COCO AP, micro- and macro- F_1 , false detections on negative frames, and synchronized batch-1 latency. For the completed YOLO runs, YOLO11n reaches $91.79 \pm 0.44\%$ mAP₅₀, $83.42 \pm 1.36\%$ macro- F_1 , and 68.0 ± 1.3 frames/s. YOLO26n instead reaches the higher mAP_{50:95} of $56.98 \pm 0.81\%$. These directions hold for every paired seed, while subject-level F_1 reveals substantially larger variation than the seed-level standard deviations. D-FINE-N results and a separate exploratory NIR training-negative exposure study are [PLACEHOLDER]. The benchmark consequently supports a bounded comparison of complete detector systems for visible cue localization, not temporal fatigue or behavior inference.

Index Terms—driver monitoring, object detection, subject-disjoint evaluation, visible driver cues, near-infrared imaging, real-time inference

I. INTRODUCTION

Driver monitoring systems operate under changes in driver identity, illumination, viewpoint, and occlusion. Whole-frame recognition can describe an activity, but it may also exploit identity or cabin context without showing the visual evidence used for a decision. Object detection provides a complementary formulation: it localizes cue-specific evidence, makes false positives and misses spatially interpretable, and measures detections on frames containing no target cue. A single-frame detector also avoids the added latency and architecture of a temporal model when real-time throughput is the experimental focus. This formulation does *not* infer fatigue, intent, duration, or any other temporal driver state.

The Driver Monitoring Dataset (DMD) provides multimodal attention and alertness recordings [1], while Drive&Act contains synchronized color, NIR, depth, and pose streams for fine-grained activities [2]. Both contain temporally correlated video and repeated observations of each driver. Evaluation can therefore be optimistic if neighboring frames or identities cross partitions, natural negative frames are removed, the test set determines the confidence threshold, or latency boundaries differ.

Recent work proposes specialized lightweight networks for in-cabin recognition and detection [3]–[6]. Our objective is different: compare complete, publicly released nano detector systems without task-specific architectural modification. The contributions are:

- a primary subject-disjoint RGB benchmark for four explicitly defined visible cues, with all sampled frames preserved and the annotation structure automatically audited;
- validation-only operating-point selection followed by a protected test, three paired RGB seeds, class- and subject-level analysis, false detections on negatives, and synchronized latency; and
- a checksum-backed reproduction workflow plus a clearly separated, one-seed NIR training-negative exposure study at 10 frames/s.

II. RELATED WORK AND EVALUATION GAP

A. Recognition Versus Localization

MobileVGG and convolution–transformer hybrids demonstrate efficient distracted-driver classification [3], [7]. PO-GUISE+ uses pose and object guidance for efficient driver-action recognition in Drive&Act NIR imagery [8]. Such whole-frame or temporal recognition tasks are not interchangeable with the present bounding-box task. Here, a correct prediction must identify both a cue class and its spatial evidence in one frame; no temporal state is estimated.

Detection-specific work modifies YOLO backbones, attention, or multiscale aggregation for dangerous-driving or fatigue cues [4]–[6]. Comparative studies also assess CNNs against transformers [9] or several YOLO generations [10]. However, differences in driver separation, retained negatives, calibration, and runtime protocol limit cross-paper ranking.

B. Released Detector Systems

DETR casts detection as bipartite set prediction [11]; RT-DETR makes the formulation practical for real-time use [12]; and D-FINE refines box-edge distributions for improved localization [13]. We compare D-FINE-N with YOLO11n and YOLO26n as shipped systems, including their native training and inference recipes. In particular, the pinned YOLO26 implementation uses end-to-end one-to-one predictions without non-maximum suppression (NMS), whereas YOLO11n uses

TABLE I
FROZEN RGB BOX SEMANTICS AND TEST SUPPORT

Cue	Bounding-box target	Test n	Median area
Yawning	Mouth	33	1.38%
Hand over mouth	Full face and covering hand	28	24.67%
Drinking	Visible hand/container-mouth interaction	56	16.09%
Phone use	Visible hand-phone interaction	497	8.37%

its conventional NMS path [14]. Thus, this is a system-level benchmark rather than an isolation of architecture alone.

III. PRIMARY RGB BENCHMARK

A. Cue Definition and Annotation

The RGB dataset contains 15,723 face crops sampled at 1 frame/s from 81 DMD recordings of 14 drivers. One primary annotator labeled the data in a single pass; no independent second-person review was established. Every sampled frame was preserved to avoid post-sampling curation, yielding 3,001 positive and 12,722 negative frames. Table I freezes the spatial ontology. Box semantics intentionally differ by cue because the visible evidence differs.

Each frame contains zero or one box. Yawning and hand-over-mouth are mutually exclusive; when a hand covers a yawn, hand-over-mouth takes precedence. Boxes are tight around only the visible target. Occluded, truncated, or out-of-frame anatomy is never estimated, although a partially occluded cue remains valid when visually identifiable. No arbitrary minimum pixel cutoff is applied at 640×640 .

An automated COCO audit found 15,723 unique images and 3,001 unique annotations, with zero duplicate IDs or file names, orphan annotations, invalid categories, nonfinite or nonpositive boxes, out-of-bounds boxes, or multi-annotation frames. The smallest annotated area was 0.81% of the frame; 139 of 159 yawning boxes occupied below 2%. This is a structural audit, not evidence of inter-annotator agreement or semantic correctness.

B. Split, Models, and Training

An exhaustive 8:3:3 subject assignment minimizes the worst relative class-distribution deviation, with root-mean-square deviation as a tie-breaker. The frozen train/validation/test partitions contain 9,087/3,423/3,213 frames. The three test subjects contribute 1,080/920/1,213 frames and 178/183/253 positives, respectively.

COCO-pretrained YOLO11n, YOLO26n [14], and D-FINE-N [13] use 640×640 input, physical batch size 8, four-step gradient accumulation (effective batch 32), 220 epochs, automatic mixed precision, and no early stopping. A fixed patch normalizes the incomplete final accumulation window by its actual sample count. Native recipes are retained: Ultralytics `optimizer=auto` with a linear schedule, and the official D-FINE AdamW/MultiStepLR recipe. Seeds {13, 37, 73} were declared in advance and no run was selected for presentation.

C. Evaluation and Uncertainty

The best model-native validation checkpoint is retained. Its confidence threshold τ^* maximizes validation micro- F_1 over $[0.01, 0.99]$, with higher precision and then higher τ as tie-breakers. At IoU 0.5, predictions are greedily matched one-to-one within class. Protected testing reports COCO mAP₅₀ and mAP_{50:95} [15], precision, recall, micro- and macro- F_1 , and

$$FD_{100} = 100 \frac{FP_{neg}}{N_{neg}}, \quad (1)$$

the number of above-threshold false detections per 100 negative frames. This is a static per-frame count, not a modeled nuisance-alert or user-experience rate; no temporal confirmation or debouncing is applied.

RGB results are the sample mean and sample standard deviation (SD) over the three seeds. Because the subject split is fixed, the SD measures optimization variability, not generalization uncertainty across held-out drivers. We therefore additionally report each test subject and use paired per-seed differences to check whether model directions are sign-consistent; no significance test is claimed for $n = 3$.

Batch-1 FP16 profiling uses an NVIDIA GeForce RTX 4060 8-GB GPU. CUDA synchronization brackets model-forward and tensor-to-final-detections intervals. Disk I/O, decoding, preprocessing, and metric computation are excluded. We report p50 tensor-to-final latency and sustained FPS. This hardware does not test claims about CPU or embedded deployment.

IV. COMPLETED RGB YOLO RESULTS

A. Aggregate and Paired-Seed Comparison

Table II summarizes the six completed RGB YOLO runs. YOLO11n has higher mAP₅₀, micro- F_1 , macro- F_1 , and throughput at every paired seed; its p50 latency is 4.67–5.17 ms lower. YOLO26n has higher mAP_{50:95} at every seed (difference 0.94–2.43 percentage points), despite lower mAP₅₀. The evidence therefore supports a system-level trade-off between coarse-IoU operating accuracy and stricter box localization, not superiority under every criterion.

B. Class, Subject, and Failure Analysis

Per-class paired differences refine the aggregate result. YOLO11n is higher for yawning at all seeds, while YOLO26n is higher for hand-over-mouth and phone use at all seeds. The sign for drinking changes by seed; the defensible conclusion is no consistent difference for that class. These observations remain descriptive because test support is only 28–56 instances for three classes (Table I).

Subject variation is more consequential. Averaged across seeds, YOLO11n micro- F_1 is 94.94%, 62.41%, and 94.72% for the three held-out subjects; YOLO26n obtains 96.11%, 54.11%, and 92.29%. Thus, the fixed-split seed SD cannot represent uncertainty over unseen drivers. Reporting only the pooled test set would hide the difficult subject.

Figure 2 shows predeclared seed-13 examples from frozen predictions. Correct detections localize the intended interaction; the shared miss occurs in glare and includes a small

TABLE II
RGB SUBJECT-DISJOINT TEST RESULTS (MEAN $\pm$ SAMPLE SD OVER SEEDS 13, 37, AND 73). LATENCY IS P50 TENSOR-TO-FINAL DETECTION; BOLD MARKS THE NUMERICAL BEST PER COLUMN, NOT STATISTICAL SIGNIFICANCE.

Model	mAP ₅₀	mAP _{50:95}	Micro- F_1	Macro- F_1	FD ₁₀₀	p50 (ms)	FPS	Params (M)	GFLOPs
YOLO11n	91.79$\pm$0.44	55.38 $\pm$ 0.41	86.60$\pm$0.49	83.42$\pm$1.36	0.63$\pm$0.04	14.30$\pm$0.20	68.0$\pm$1.3	2.59	6.44
YOLO26n	89.28 $\pm$ 0.99	56.98$\pm$0.81	84.44 $\pm$ 1.14	79.39 $\pm$ 1.49	0.67 $\pm$ 0.06	19.21 $\pm$ 0.09	51.7 $\pm$ 0.2	2.51	5.78
D-FINE-N	[PLACEHOLDER]								

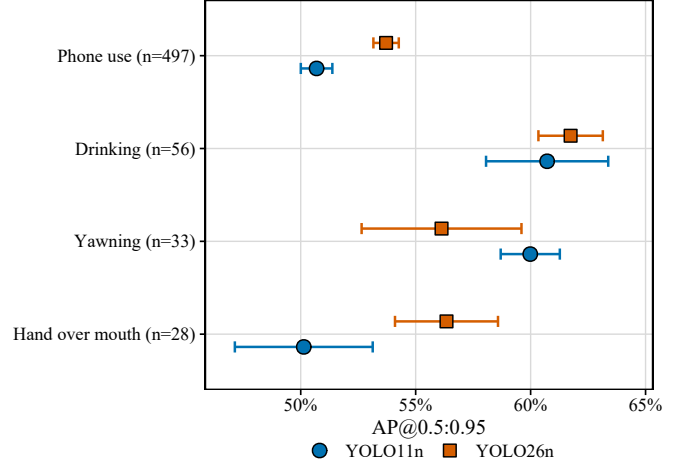
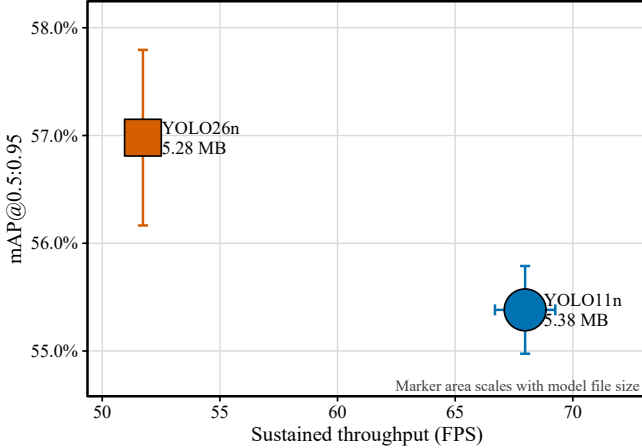


Fig. 1. Completed RGB YOLO results. Error bars are sample SD across seeds 13, 37, and 73. In (a), marker area is proportional to the serialized inference artifact size. Generated from frozen result files using R/ggplot2.

drinking target. More importantly, one nominal negative appears visually cue-like. It may reflect a difficult ontology boundary or a missed label, so the structural annotation audit alone cannot establish semantic validity. This example is retained because it is informative about the benchmark’s principal annotation risk.

V. EXPLORATORY NIR TRAINING-NEGATIVE EXPOSURE STUDY

NIR is a separate secondary experiment, not a co-equal spectral track or a paired domain-shift test. Its source pool contains 3,786 one-second front-top active-NIR (850 nm) Drive&Act snippets from 15 drivers. Twenty-three unilluminated task snippets are excluded before splitting, leaving 3,763. Drinking and phone-use tracklets provide the two positive classes; 10 frames per snippet are sampled at exactly 10 frames/s, cropped to the driver region, resized to 640 $\times$ 640, and interpolated at 0.1-s intervals. Unlike RGB, NIR therefore has dataset-specific frame exclusion and interpolated annotations.

A fixed 9:3:3 subject partition is shared across every model and condition. Validation and test contain identical ordered examples. Training always contains 270 positive snippets; conditions contain either 540 or 1,620 negative snippets (nominal positive:negative ratios 1:2 and 1:6). The smaller negative pool is a subject-stratified seed-13 subset of the larger. With epochs held fixed, 1:6 also has three times the unique negatives and approximately three times the optimization steps. The experiment is therefore a *training-negative exposure study*, not a causal isolation of class ratio under matched training signal.

TABLE III
EXPLORATORY NIR TEST RESULTS AT SEED 13

Model	Neg.	AP	Frame	Snippet	FP snippets	p50
			macro- F_1	macro- F_1	per 100	
YOLO11n	1:2		[PLACEHOLDER]			
YOLO11n	1:6		[PLACEHOLDER]			
YOLO26n	1:2		[PLACEHOLDER]			
YOLO26n	1:6		[PLACEHOLDER]			
D-FINE-N	1:2		[PLACEHOLDER]			
D-FINE-N	1:6		[PLACEHOLDER]			

All six jobs use seed 13. Frame-level metrics follow the RGB protocol. Because the ten frames within a one-second snippet are near-duplicates, snippet-level operating metrics treat each snippet as one unit: a class is recovered if any frame contains a correctly matched prediction, repeated detections count once, and a negative snippet is false positive if any frame contains an above-threshold detection. Table III will report both granularities after the protected evaluations.

VI. LIMITATIONS

The RGB sample contains 14 drivers and only three held-out test drivers, so subject-disjoint evaluation reduces identity leakage but does not establish population-level generalization. Seed SD on this fixed split measures optimization variability only. One annotator completed one pass without independent review; the audit verifies file and geometric constraints but not semantic agreement, and the qualitative analysis exposes at least one ambiguous or potentially missed label. Frames

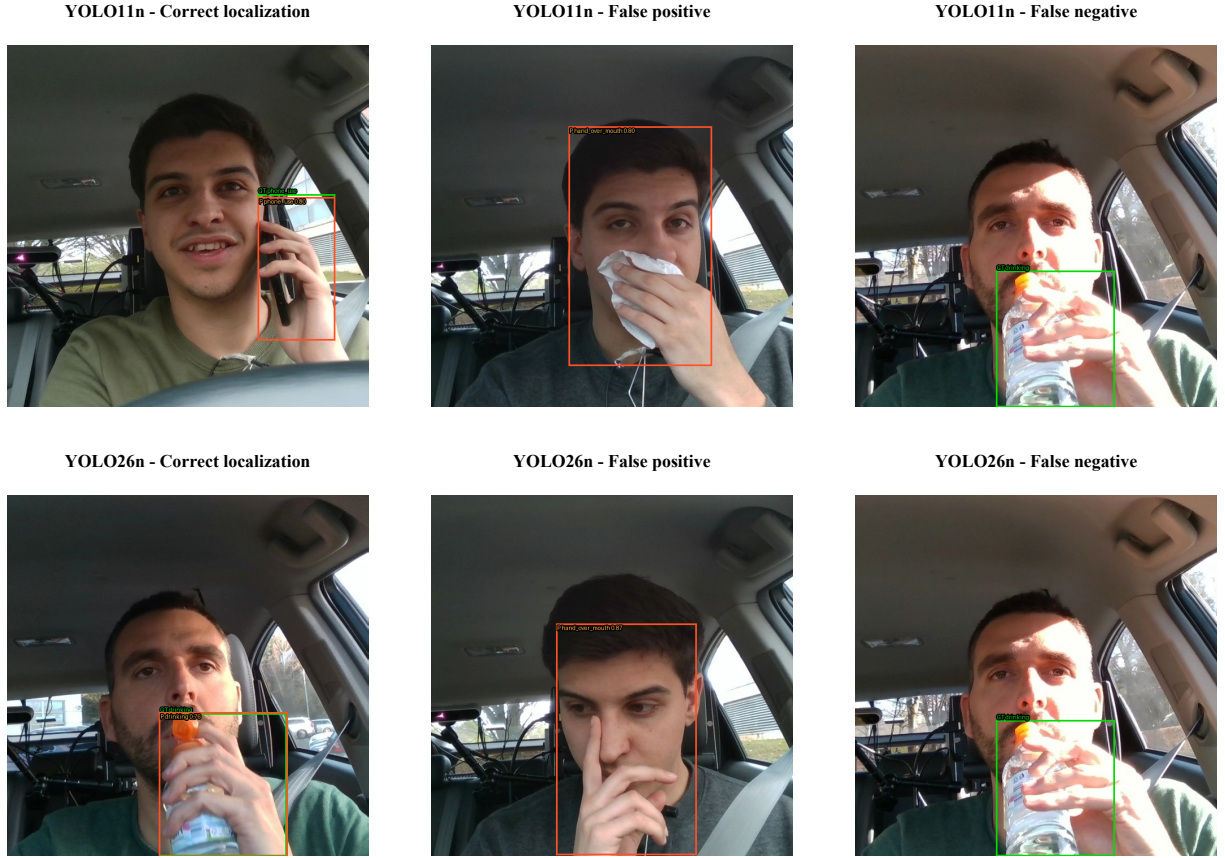


Fig. 2. Representative seed-13 RGB outcomes. Green boxes denote ground truth and orange boxes denote predictions at the validation-selected threshold. Rows are complete detector systems; columns show a true positive, a false detection on a nominal negative, and a false negative. The cue-like nominal negative motivates independent semantic review in future versions. Generated with R/ggplot2.

from the same recording remain correlated, and no clip-level uncertainty is claimed for RGB. The single-frame task does not model temporal fatigue or alert logic.

The NIR experiment differs in dataset, ontology, annotation source, sampling rate, and preprocessing, uses one seed, and confounds negative diversity with training exposure. Its results cannot be compared with RGB as a controlled modality effect. Finally, all models retain system-specific optimization and postprocessing, while RTX 4060 latency excludes preprocessing and is not an embedded-device measurement. These constraints limit conclusions to the declared systems, data, and metrics.

VII. CONCLUSION

The completed RGB comparison shows a reproducible trade-off: YOLO11n provides the stronger validation-calibrated operating point and throughput, while YOLO26n provides higher strict-IoU AP through its distinct end-to-end system. Paired seeds support those aggregate directions, but class and subject analyses prevent overgeneralization. The final three-model RGB ranking and the separate NIR exposure findings are [PLACEHOLDER]. The resulting contribution

is a transparent benchmark for spatial localization of visible driver cues, not a claim of temporal driver-state inference.

DATA AND CODE AVAILABILITY

Authored annotations, subject partitions, checksums, frozen YOLO test predictions, evaluation code, and publication artifacts are available at <https://github.com/IamOumarIbrahim/lightweight-driver-behavior-detection>. DMD and Drive&Act media remain subject to their original licenses; authorized users can reconstruct the tracks with the published scripts. Trained checkpoints remain local.

REFERENCES

- [1] J. D. Ortega, N. Kose, P. Cañas, M.-A. Chao, A. Unnervik, M. Nieto, O. Otaegui, and L. Salgado, “DMD: A large-scale multi-modal driver monitoring dataset for attention and alertness analysis,” in *Computer Vision – ECCV 2020 Workshops*. Springer, 2020, pp. 387–405.
- [2] M. Martin, A. Roitberg, M. Haurilet, M. Horne, S. Reiß, M. Voit, and R. Stiefelhagen, “Drive&Act: A multi-modal dataset for fine-grained driver behavior recognition in autonomous vehicles,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 2801–2810.
- [3] B. Baheti, S. Talbar, and S. Gajre, “Towards computationally efficient and realtime distracted driver detection with MobileVGG network,” *IEEE Transactions on Intelligent Vehicles*, vol. 5, no. 4, pp. 565–574, 2020.

- [4] C. Wang, M. Lin, L. Shao, and J. Xiang, "MF-YOLO: A lightweight method for real-time dangerous driving behavior detection," *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–13, 2024.
- [5] R. Li, C. Yu, X. Qin, X. An, J. Zhao, W. Chuai, and B. Liu, "YOLO-SGC: A dangerous driving behavior detection method with multiscale spatial-channel feature aggregation," *IEEE Sensors Journal*, vol. 24, no. 21, pp. 36 044–36 056, 2024.
- [6] Z. Jin and L. Dong, "YOLO11-CR: A lightweight convolution-and-attention framework for accurate fatigue driving detection," 2025, arXiv:2508.13205.
- [7] Z. Li, X. Zhao, F. Wu, D. Chen, and C. Wang, "A lightweight and efficient distracted driver detection model fusing convolutional neural network and vision transformer," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 12, pp. 19 962–19 978, 2024.
- [8] R. Pizarro, R. Valle, R. Barea, J. M. Buenaposada, L. Baumela, and L. M. Bergasa, "PO-GUISE+: Pose and object guided transformer token selection for efficient driver action recognition," 2024, arXiv:2407.13750.
- [9] H. V. Koay, J. H. Chuah, and C.-O. Chow, "Convolutional neural network or vision transformer? benchmarking various machine learning models for distracted driver detection," in *2021 IEEE Region 10 Conference (TENCON)*, 2021, pp. 417–421.
- [10] D. Herath, C. Abeyrathne, and P. Jayaweera, "Vision-based driver drowsiness monitoring: Comparative analysis of YOLOv5–v11 models," 2025, arXiv:2509.17498.
- [11] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Computer Vision – ECCV 2020*. Springer, 2020, pp. 213–229.
- [12] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, "DETRs beat YOLOs on real-time object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 16 965–16 974.
- [13] Y. Peng, W. Lv, Y. Hu, X. Li, Z. Sun, Z. He, H. Guo, Y. Cai, J. Chen, Y. Zhang, and Y. Zhang, "D-FINE: Redefine regression task in DETRs as fine-grained distribution refinement," in *The Thirteenth International Conference on Learning Representations*, 2025.
- [14] Ultralytics, "YOLO26: End-to-end NMS-free detection," <https://docs.ultralytics.com/models/yolo26/>, 2026, accessed: Aug. 25, 2026.
- [15] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Computer Vision – ECCV 2014*. Springer, 2014, pp. 740–755.